

ULTRA GYM: Transforming Clinical 3D Ultrasound Volumes into Dynamic, Multi-Task Training Data

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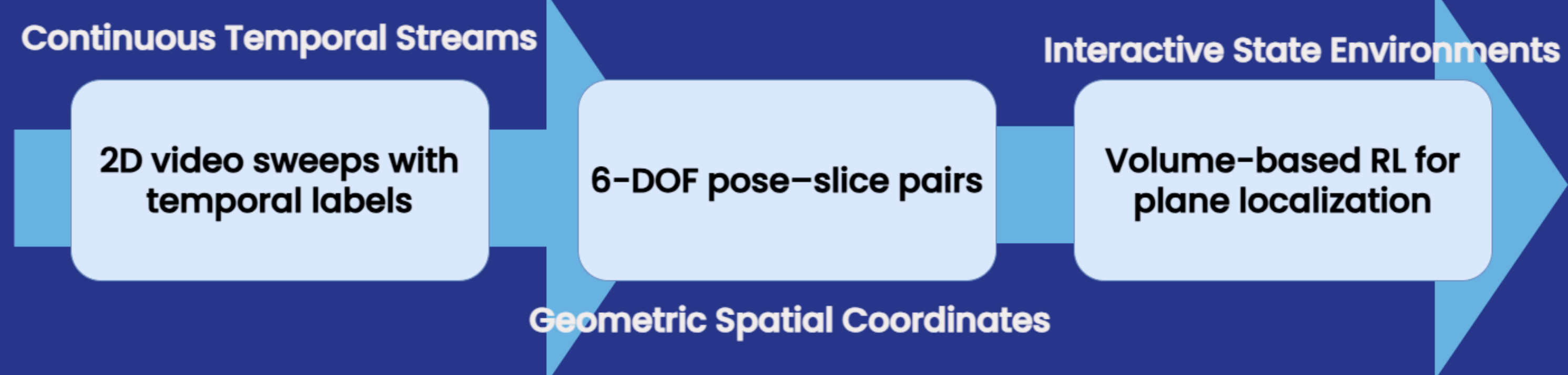
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MOTIVATION

- 3D ultrasound volumes are rich, but largely underutilized.
- Cross-modality synthesis loses ultrasound-specific physics.
- Existing tools are often limited to single tasks

APPROACH



One clinical volume → 3 training-ready outputs

STANDARD PLANE DETECTION

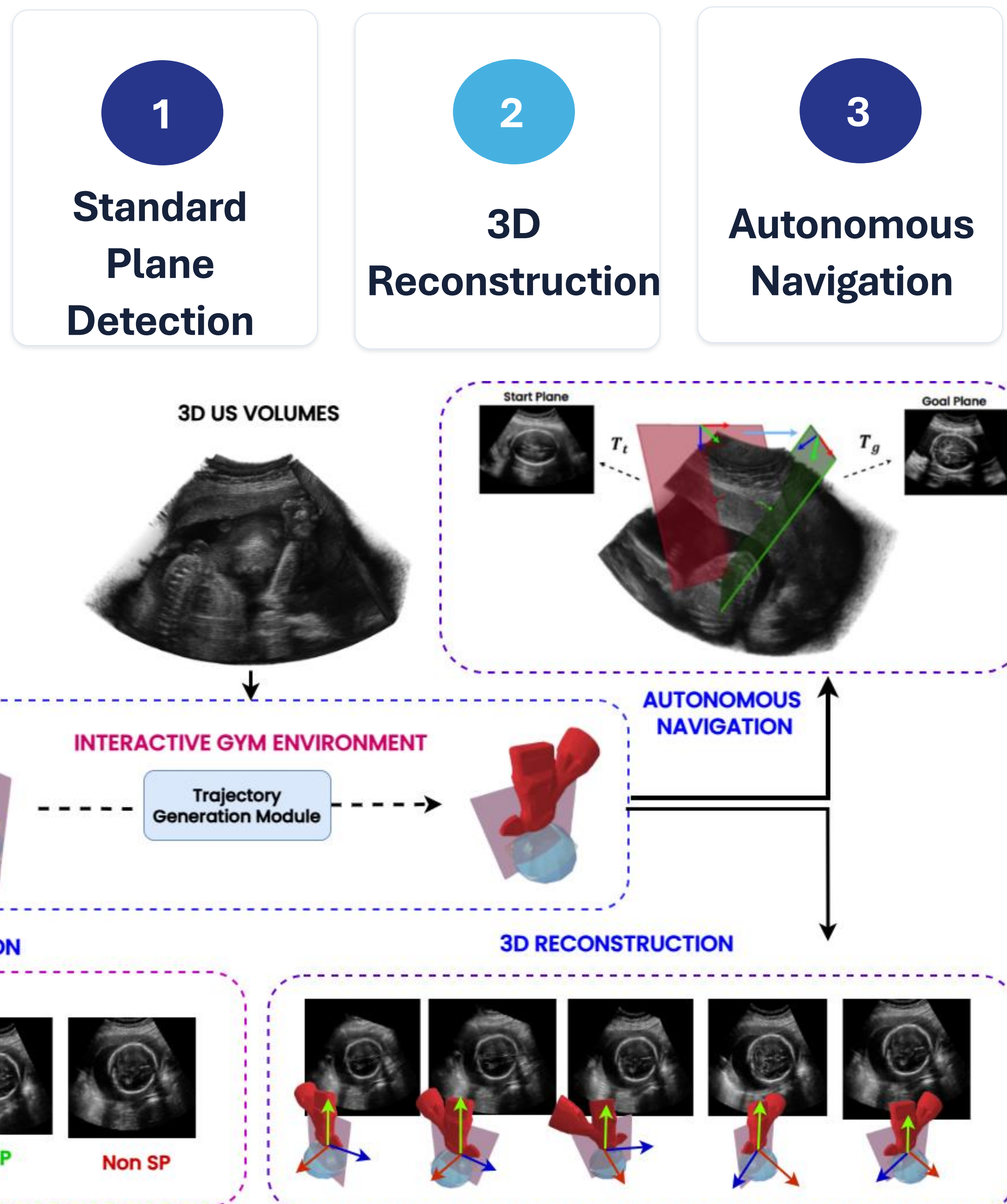
- Randomized probe starts generate diverse initial trajectories.
- Sweep-through and overshoot capture realistic search behavior
- Proximity-based labels quantify distance to anatomical targets.
- ϵ -sphere micro-trajectories capture fine-grained local movements.

3D RECONSTRUCTION

- Dense pre-planned sweeps ensure comprehensive coverage
- Absolute coordinate control enables precise probe positioning.
- Perfectly paired outputs provide aligned training data.

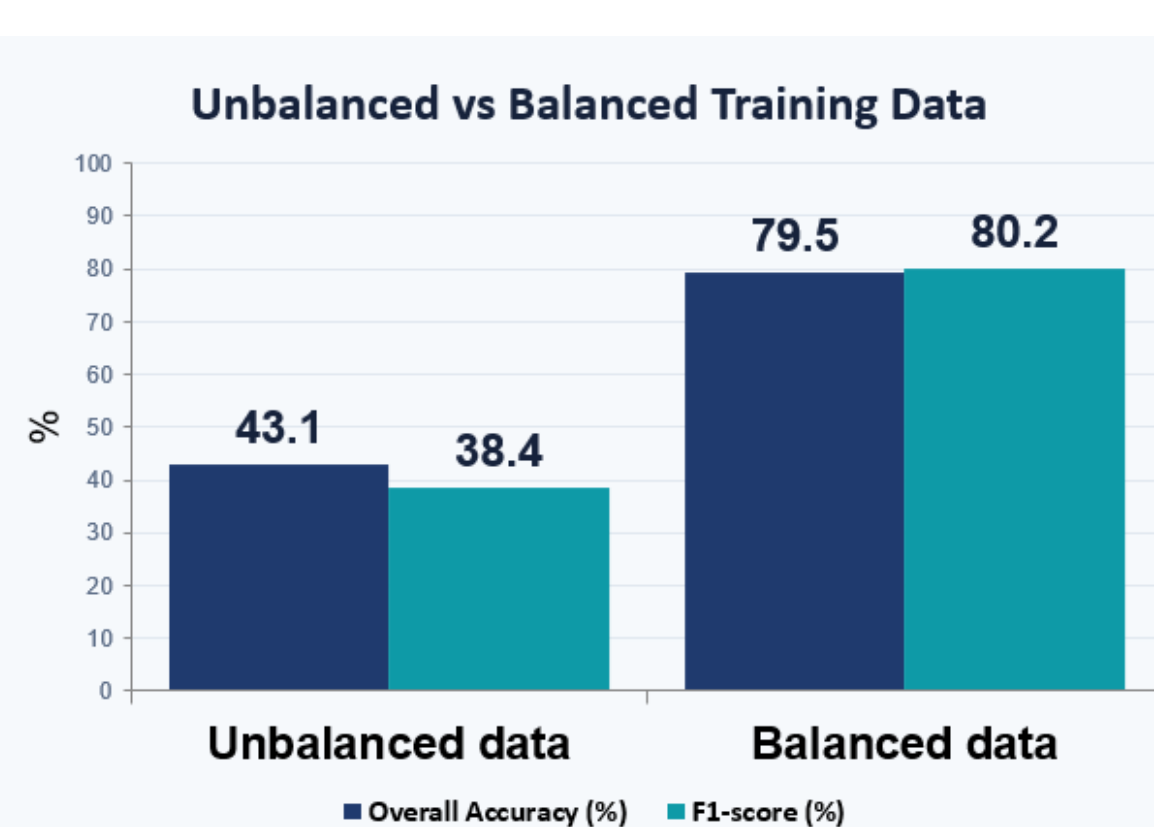
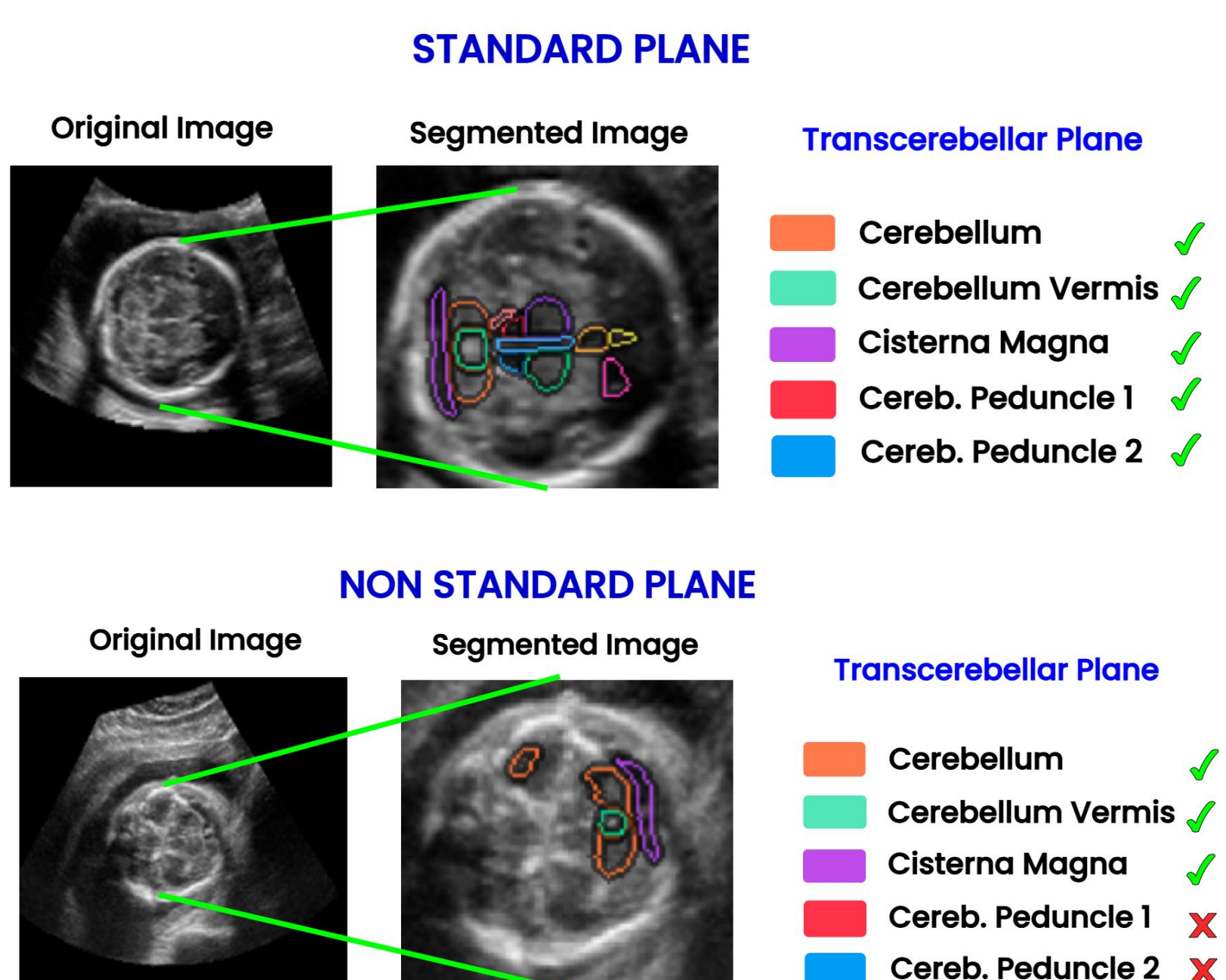
AUTONOMOUS NAVIGATION

- Localization is framed as an MDP for goal-directed probe navigation



RESULTS

STANDARD PLANE UNDERSTANDING

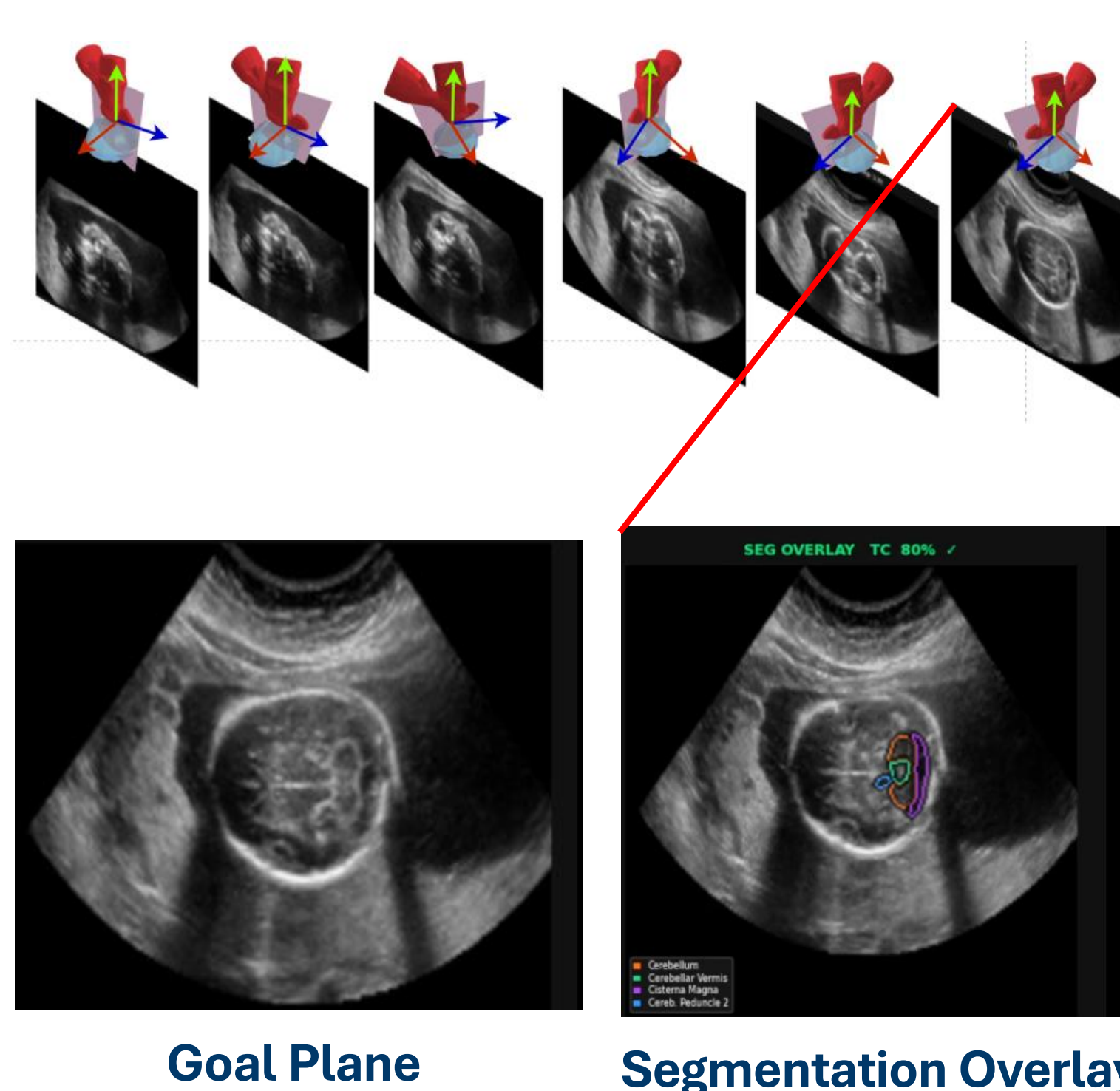


SP / Non-SP
ACCURACY
(Balanced)
83.93%
Standard-Plane
accuracy
71.29%
Non-Standard-
Plane accuracy

3D RECONSTRUCTION

Metric	Pairwise CNN	DCL-Net Baseline
Local PDE Avg. (mm)	0.14	0.12
Local PDE Range (mm)	0.03 – 0.27	0.04 – 0.22
Final Drift Avg. (mm)	10.80	8.47
Final Drift Range (mm)	0.51 – 42.00	0.80 – 27.14

AUTONOMOUS NAVIGATION



78 ± 13 Average episode Length (Steps)
84% Navigation Success Rate

KEY TAKEAWAYS

- ✓ Native acoustic fidelity through direct clinical-volume slicing
- ✓ Multi-task data generation from a single clinical archive.
- ✓ Modular framework for broader medical-imaging pipelines.
- ✓ Open-source dataset, code, and videos for the community.

Ultra GYM presents a **scalable framework** for transforming clinical 3D ultrasound volumes into diverse, **task-ready data** for medical AI.

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